

## LESSON 6-7 • SESSION 2 • PRACTICE • FORM B

# Multiples and Least Common Multiple

Learning target – I can list multiples of two numbers and find their least common multiple. • 6.NS.B.4

**REMEMBER**

- A **multiple** is what you get when you skip-count by a number: 6, 12, 18, 24, ...
- The **least common multiple (LCM)** is the smallest multiple above 0 that two numbers share.
- Watch out:  $8 \times 10 = 80$  is a common multiple, but it is **not always the least**.

**WORKED EXAMPLE**

A library van visits every 8 days and an art cart every 10 days. Both visit today. When do both visit again?

**Step 1** Multiples of 8: 8, 16, 24, 32, 40.

**Step 2** Multiples of 10: 10, 20, 30, 40.

**Step 3** The first number on both lists is 40.

**Answer** LCM = 40, so in 40 days

**WORDS** multiple (*múltiplo*) • common multiple (*múltiplo común*) • least common multiple (*mínimo común múltiplo*)

**PRACTICE***show your work • label units*

1. Find the LCM of each pair.

a. 6 and 15 \_\_\_\_\_

b. 10 and 25 \_\_\_\_\_

c. 3 and 8 \_\_\_\_\_

d. 12 and 16 \_\_\_\_\_

2. **Two Ferries.** A river ferry reaches a dock every 12 minutes. A harbor ferry reaches the same dock every 18 minutes. Both are at the dock now.

a. Multiples of 12: \_\_\_\_\_ Multiples of 18: \_\_\_\_\_

b. Minutes until both ferries are there again: \_\_\_\_\_ min

3. **Juice and Straws.** Juice boxes come in packs of 6. Straws come in sleeves of 8. A store wants the same number of juice boxes and straws.

**Hint** Skip-count by 8 and stop at a multiple of 6.

a. Fewest juice boxes and fewest straws: \_\_\_\_\_

b. Packs of juice \_\_\_\_ Sleeves of straws \_\_\_\_

Show your work.

4. **Club Days.** Art club meets every 4 days. Chess club meets every 7 days. Both met today. In how many days will both meet again?

**Hint** List the meeting days for each club.

Show your work.

**Answer** \_\_\_\_\_ days

5. **Tess's Shortcut.** Tess says the LCM of 18 and 24 is 432 because  $18 \times 24 = 432$ . Is she right? Give the LCM.

**Hint** Skip-count by 24 and test each number.

Show your work.

**LCM** \_\_\_\_\_

- 6. Two Machines.** Machine A stamps every 20 seconds. Machine B stamps every 35 seconds. They both stamped just now.
- Hint** Divide the LCM by each machine's time.
- a.** Seconds until they stamp together again: \_\_\_\_\_ s
- b.** Stamps by A \_\_\_\_ Stamps by B \_\_\_\_ (up to that time)

Show your work.

- 7. Three Lists.** Find the least common multiple of 3, 4, and 6.

**Hint** Skip-count by 6 and look for a multiple of 4.

Show your work.

LCM \_\_\_\_\_

- 8. Two Chimes.** One chime sounds every 8 minutes. Another sounds every 20 minutes. Both sound at 8:00 a.m.
- a.** Find the LCM of 8 and 20. \_\_\_\_\_
- b.** When do both sound together next? \_\_\_\_\_
- c.** How many times do they sound together after 8:00 and up to 10:00 a.m.? \_\_\_\_\_

Show your work.

### FIND THE ERROR

explain the mistake, then fix it

- 9. Elena's Work.** Elena listed multiples of 6 and of 8. She said the LCM is 48, the second number that appeared on both lists. Explain her mistake. Then give the correct LCM.

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### TEST PRACTICE

choose the best answer

- 10.** The LCM of 3 and 5 is –
- A** 1                      **B** 8                      **C** 15                      **D** 30
- 11.** Which number is a multiple of 9?
- A** 3                      **B** 19                      **C** 29                      **D** 36

### SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|                       |                       |                        |                       |                       |                       |                      |                         |                       |                        |
|-----------------------|-----------------------|------------------------|-----------------------|-----------------------|-----------------------|----------------------|-------------------------|-----------------------|------------------------|
| <b>24</b><br><b>U</b> | <b>40</b><br><b>N</b> | <b>140</b><br><b>I</b> | <b>36</b><br><b>Q</b> | <b>12</b><br><b>E</b> | <b>72</b><br><b>T</b> | <b>6</b><br><b>X</b> | <b>8:40</b><br><b>T</b> | <b>28</b><br><b>O</b> | <b>432</b><br><b>Z</b> |
| #2b                   | #3a                   | #4                     | #5                    | #6a                   | #7                    | #8a                  | #8b                     |                       |                        |

### CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_